

Tracking Competing Narratives in the Russia–Ukraine Conflict Using Stance-Aware Epidemiological Models

Nitin Agarwal^{1,2} and Muhammad Hassam Aslam Khan¹

¹ COSMOS Research Center, University of Arkansas–Little Rock, USA

² International Computer Science Institute, University of California, Berkeley, USA

{nxagarwal,mkhan5}@ualr.edu

Abstract. Social media platforms play an important role in shaping and spreading political narratives during armed conflicts. This study analyzes approximately 1.6 million Twitter posts related to the Russia–Ukraine war using an LLM-based pipeline to identify major narrative themes and classify posts as Pro-Ukraine, Pro-Russia, or Neutral. The study asks which stance streams are most strongly associated with each narrative’s diffusion. We construct a daily first-time unique-user adoption series and examine narrative diffusion using two epidemiological models. A calibrated SEIZ model captures aggregate narrative spread, while a stance-stratified $SEI_A I_D Z$ model estimates separate transmission patterns for the Pro-Ukraine, Pro-Russia, and Neutral streams within a shared susceptible population. Model reliability is evaluated using multi-start optimization, bootstrap uncertainty, identifiability diagnostics, and temporal holdout validation. The proposed framework provides a systematic way to compare aggregate narrative diffusion with stance-specific dynamics and offers a reliability-aware approach to studying competing political narratives in online conflict communication.

Keywords: Russia–Ukraine conflict · Narrative diffusion · Stance detection · Epidemiological modeling · Social media analysis

1 Introduction

The Russia–Ukraine conflict has generated a large volume of online discussion used to share news, promote competing interpretations, and influence opinion. Telegram and Twitter/X have been central to war-related communication, propaganda, and misinformation [6, 5, 14], and because the discourse is large, fast-moving, and contested, computational methods are needed to organize it and study how narratives spread.

Individual posts are short and hard to interpret in isolation. LDA and BERTopic [7, 11] provide topic-level organization but struggle when opposing sides share vocabulary, so we use an LLM-based clustering approach to identify narrative themes. Clustering alone does not reveal which position a post supports; we

therefore add stance classification (Pro-Ukraine, Pro-Russia, Neutral), which is distinct from sentiment because opposing sides may express similar emotions [4]. We then model diffusion with a calibrated SEIZ baseline and a stance-stratified $SEI_A I_D Z$ model over a common observation window.

The central motivation is to move beyond identifying what narratives exist and examine how they diffuse over time. For each narrative, we ask which stance stream is most strongly associated with its diffusion. This allows us to distinguish narratives that are broadly visible from those whose spread is concentrated within a particular political stance.

The corpus covers the early months of the Russia–Ukraine conflict through May 17, 2022. The diffusion models are estimated over the continuous analysis period from March 23 to May 17, 2022. This paper contributes: (1) an LLM-based pipeline for conflict-related narrative themes; (2) a user-level diffusion measure based on first observed adoption; (3) an aggregate SEIZ baseline; and (4) a stance-stratified $SEI_A I_D Z$ model with separate latent exposure states and separate Pro-Ukraine, Pro-Russia, and Neutral transmission parameters within each narrative. Both model families are evaluated using multistart optimization, bootstrap uncertainty, identifiability checks, and temporal holdout validation.

2 Related Work

The conflict has been widely studied as information warfare. Prior work shows that pro-Kremlin and anti-Kremlin communities used Telegram to spread and defend competing narratives [6], that Twitter/X enabled coordinated messaging and network-level amplification [5], and that war narratives cross languages and communities [14]. A recurring finding—that opposing streams differ in structure and amplification, not only content—motivates modeling competing positions separately rather than as one undifferentiated theme.

Organizing such data requires clustering. LDA [7] and BERTopic [11] extract topic structure but falter when opposing sides share vocabulary; LLM clustering incorporates framing and intent, and LLMs can match human annotators at scale [9], provided a frozen taxonomy is applied consistently, which our Phase A consensus enforces. Stance detection, distinct from sentiment [4], has been applied to Brexit, COVID-19, and elections but rarely to active conflict. On the modeling side, the skeptical compartment in SEIZ captures selective engagement [15]; related models address narrative diffusion [3] and stance-aware contagion [1, 2], and Gurung et al. show stance-aware models can expose heterogeneity aggregate models miss [12]. We extend this literature by estimating stance-specific diffusion within each narrative. The three stance streams retain separate latent exposure states and transmission parameters, but share the susceptible population and post-adoption transition rates. This design supports within-narrative comparison while limiting the number of free parameters and withholding conclusions when numerical, bootstrap, or held-out checks fail.

3 Methodology

The pipeline has four stages (Fig. 1): clustering; construction of user-level first-adoption series; aggregate SEIZ modeling; and stance-stratified $SEI_A I_D Z$ modeling.

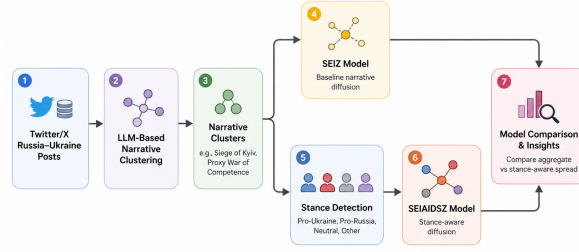


Fig. 1. Pipeline for aggregate and stance-aware narrative diffusion analysis.

Narrative Clustering. Stratified samples are passed through multiple LLM agents that draft and refine narrative labels, resolve overlap, and define boundary rules. The resulting taxonomy is frozen and applied in parallel to the full corpus, producing a cluster identifier, confidence score, and rationale for each post; low-confidence and overflow cases are reviewed without modifying the taxonomy. Prompts, taxonomy, model configuration, and output schema are retained for reproducibility. A companion study currently under review reports 0.89 accuracy, 0.88 macro-F1, and substantial inter-annotator agreement ($\alpha = 0.793$) for the underlying clustering methodology.

Analysis Window and First-Adoption Series. The diffusion analysis uses the continuous period from March 23–May 17, 2022 (56 days). Observations from March 18–21 are used to initialize the model, while March 22 is treated as unavailable rather than as zero activity. The observation unit is a user’s first observed adoption of a narrative. Daily incidence therefore counts new first adopters rather than posts, and cumulative series are obtained by summing daily incidence. The modeled population is the distinct-user collection population, $N = 464,268$.

Calibrated SEIZ. SEIZ partitions users into Susceptible (S), Exposed (E), Infected (I , adopters), and Skeptical (Z) [13]:

$$\frac{dS}{dt} = -\beta S \frac{I}{N} - bS \frac{Z}{N}, \quad (1)$$

$$\frac{dE}{dt} = (1-p)\beta S \frac{I}{N} + (1-l)bS \frac{Z}{N} - \rho E \frac{I}{N} - \epsilon E, \quad (2)$$

$$\frac{dI}{dt} = p\beta S \frac{I}{N} + \rho E \frac{I}{N} + \epsilon E, \quad \frac{dZ}{dt} = lbS \frac{Z}{N}. \quad (3)$$

Table 1. Human and cross-model validation of stance classification on 600 posts

Panel A: Gemma performance against human labels			
Stance	Precision	Recall	F1
Pro-Ukraine	0.87	0.89	0.88
Pro-Russia	0.83	0.80	0.81
Neutral	0.79	0.81	0.80
Macro Avg.	0.83	0.83	0.83
Panel B: Agreement with original Gemma labels			
Model	Agreement	Cohen's κ	Macro F1
GPT-5.6	0.802	0.656	0.789
Claude Opus 4.8	0.827	0.697	0.811

Here, β and b are the $S \rightarrow I$ and $S \rightarrow Z$ contact rates, ρ is the contact-driven $E \rightarrow I$ rate, and ϵ is the exposed-user self-adoption rate. Because the observed series cannot identify all parameters, we estimate only β and ϵ , fixing $p = 0.50$, $l = 0.75$, $\rho = 0.05$, and $b/\beta = 0.50$ following [13]. Initial exposed and skeptical counts are varied across three prespecified scenarios. Estimation uses a negative-binomial likelihood, multistart optimization, bootstrap uncertainty, and an 11-day temporal holdout.

Stance Detection. Posts were classified as Pro-Ukraine, Pro-Russia, or Neutral using `google/gemma-4-31B-it` [8], whose open weights permit local deployment [10]. On 600 manually labeled posts, Gemma achieved a macro F1 of 0.83 (Table 1). Cross-model checks with GPT-5.6 Sol and Claude Opus 4.8 yielded Cohen's κ values of 0.656 and 0.697, respectively.

Stance-Stratified $SEI_A I_D Z$. For each narrative, we jointly fit the daily Pro-Ukraine, Pro-Russia, and Neutral first-adopter incidence series [1, 12]. The model uses one shared susceptible population and three stance-specific exposed states, E_A , E_D , and E_Z :

$$\begin{aligned}
\frac{dS}{dt} &= -\beta_A S \frac{I_A}{N} - \beta_D S \frac{I_D}{N} - \beta_Z S \frac{Z}{N} - \eta \frac{S}{N}, \\
\frac{dE_s}{dt} &= \beta_s S \frac{I_s}{N} + w_s \eta \frac{S}{N} - (\psi + \mu) E_s, \quad s \in \{A, D\}, \\
\frac{dE_Z}{dt} &= \beta_Z S \frac{Z}{N} + w_Z \eta \frac{S}{N} - (\psi + \mu) E_Z, \\
\frac{dI_A}{dt} &= \psi E_A + \delta_D I_D + \nu_A Z - (\delta_A + \gamma_A + \mu) I_A, \\
\frac{dI_D}{dt} &= \psi E_D + \delta_A I_A + \nu_D Z - (\delta_D + \gamma_D + \mu) I_D, \\
\frac{dZ}{dt} &= \psi E_Z + \gamma_A I_A + \gamma_D I_D - (\nu_A + \nu_D + \mu) Z.
\end{aligned} \tag{4}$$

Here, I_A , I_D , and Z represent first observed Pro-Ukraine, Pro-Russia, and Neutral adoption, respectively, and the observed incidence streams are ψE_A , ψE_D , and ψE_Z . Initial exposure is allocated using pre-window stance shares.

The estimated parameters are β_A , β_D , β_Z , the shared transition rate ψ , and the initial exposed population E_0 ; the remaining transition rates are fixed. The model is fitted to daily incidence using a negative-binomial likelihood, multistart optimization, and bootstrap uncertainty. The coupled reproduction number is $R_0 = \rho(FV^{-1})$, where $\rho(\cdot)$ denotes the spectral radius. Counterfactual stance $R_{0,s}$ values are obtained by setting the other two stance transmission parameters to zero.

Reliability Criteria. For both model families we examine multistart agreement, sensitivity-matrix rank, condition number, bootstrap convergence, boundary solutions, incidence fit, and holdout performance. For SEIZ we additionally require stability of β and ϵ across the three sensitivity scenarios. A stance estimate is interpreted only when its stance-specific diagnostics and holdout comparison pass. A formal Pro-Ukraine–Pro-Russia leadership conclusion is reported only when both geopolitical stance estimates are reliable within the same narrative. Coupled narrative-level R_0 values use a separate overall reliability flag. All models are fitted to daily incidence; cumulative curves are derived by summation and are not fitted directly.

4 Results

Among the 1,566,962 collected posts, 1,547,159 were grouped into the eight narrative clusters reported in Table 2. All compartmental analyses were conducted using the continuous 56-day window from March 23 to May 17, 2022.

Narrative Clusters. The largest cluster, *The Global Frontline of Democracy* (766,364 posts), frames Ukraine as democracy’s frontline; the second, *The Anatomy of Atrocity & Accountability* (242,306), addresses civilian suffering and accountability. The remaining six range from 178,909 to 12,048 posts. Post counts describe the full corpus, whereas model parameters use first-adopter activity in the covered window.

SEIZ. The aggregate SEIZ estimates vary substantially across narratives (Table 2). *The Invisible Front* has the largest numerical β (0.0278), followed by *The Diplomacy Trap* (0.0237), while *The War for Truth* has the smallest (0.0133), a roughly 2.1-fold range.

In the SEIZ model, a larger β indicates faster estimated contact-mediated transmission, but does not necessarily imply the largest final audience, which also depends on the initial exposed population, ϵ , the available susceptible population, and the observation period. Although *The Invisible Front* has the largest numerical β , it does not meet all predefined reliability criteria and is therefore interpreted descriptively; its ϵ confidence interval also reaches the admissible parameter bounds.

Among the reliability-qualified calibrated SEIZ fits, *The Diplomacy Trap & The Cost of Peace* has the larger contact-transmission estimate and outperforms

Table 2. Narrative clusters and calibrated SEIZ results

ID	Narrative	Posts	Description	β [95% CI]	ϵ [95% CI]	Cum. Err.	Holdout ODE/Base
C003	The Global Frontline of Democracy	766,364	Ukraine as democracy's frontline against authoritarian aggression; Western unity and military support.	0.0216 [0.0172, 0.0272]	0.1725 [0.0530, 0.5683]	0.1838	1130.57 / 774.34
C002	The Anatomy of Atrocity & Accountability	242,306	Civilian suffering, war crimes, mass graves, legal and moral accountability.	0.0143 [0.0115, 0.0192]	0.1172 [0.0392, 0.3528]	0.3303	746.24 / 163.52
C001	The Looming Storm & The Grinding Front	178,909	Shift from invasion warnings to prolonged attrition in the east.	0.0188 [0.0142, 0.0248]	0.2131 [0.0409, 0.0165]	0.3254	449.84 / 223.32
C005	The Orwellian Fortress	163,928	Russia as an information fortress where propaganda conceals war costs.	0.0175 [0.0133, 0.0235]	0.1880 [0.0423, 0.7650]	0.2709	697.67 / 422.26
C008	The War for Truth	106,264	The conflict as documented evidence versus Kremlin disinformation.	0.0133 [0.0097, 0.0177]	0.1599 [0.0387, 0.6457]	0.3144	339.82 / 262.50
C006	The Diplomacy Trap & The Cost of Peace	39,599	Negotiations as a strategic battlefield of delays and concessions.	0.0237 [0.0187, 0.0326]	0.1736 [0.0431, 1.4468]	0.2181	421.29 / 438.93
C004	The Bastion of Freedom	37,741	Ukrainian resilience and unity in defense of freedom.	0.0180 [0.0121, 0.0289]	0.5594 [†] [0.0200, 1.5000]	0.1324	200.44 / 109.93
C007	The Invisible Front (Cyber-Kinetic)	12,048	Hybrid conflict where cyberattacks support battlefield operations.	0.0278 [0.0195, 0.0485]	0.1741 [†] [0.0215, 1.5000]	0.2900	243.16 / 225.59

Notes: Counts refer to 1,547,159 assigned posts; intervals are 95% bootstrap CIs (200 replications). Cum. Err. is relative L_2 error, and Holdout is ODE versus seven-day moving-average RMSE over 11 days. [†]Boundary-reaching ϵ estimates are not interpreted. Bold entries satisfy all SEIZ reliability criteria.

the seven-day moving-average baseline in all three prespecified sensitivity scenarios, with a central-scenario holdout RMSE of 421.29 compared with 438.93. *The War for Truth* outperforms the baseline in two of the three scenarios, with a central-scenario RMSE of 339.82 compared with 362.50. However, aggregate SEIZ does not identify how much of the observed diffusion is associated with each stance stream, motivating the stance-stratified analysis that follows.

Stance-Aware $SEI_A I_D Z$. The largest numerical coupled estimate occurs for *The Invisible Front*, with $R_0 = 1.588$ ([1.143, 2.576]). This result is associated primarily with the Pro-Ukraine stream, which has the largest counterfactual stance R_0 in the table (1.327), compared with 0.540 for Neutral and 0.009 for Pro-Russia. The second-largest coupled estimate occurs for *The Global Frontline of Democracy* ($R_0 = 1.505$, [1.187, 1.821]), where the Pro-Ukraine ($R_{0,PU} = 0.868$) and Neutral ($R_{0,N} = 0.858$) streams have similarly large counterfactual estimates.

Among the Pro-Russia streams, *The Diplomacy Trap & The Cost of Peace* has the largest counterfactual estimate ($R_{0,PR} = 0.232$), although it remains below the corresponding Pro-Ukraine (0.823) and Neutral (0.624) estimates.

These R_0 estimates describe diffusion potential near the diffusion-free state and should not be interpreted as evidence that observed incidence increased throughout the analysis window.

Across narratives, all 40 multistart optimizations and 200 bootstrap replications completed, with bootstrap convergence of 1.00. Condition numbers ranged from 3.915 to 11.370; only the *Atrocity & Accountability* fit placed E_0 on its admissible boundary, and E_0 is therefore not interpreted substantively.

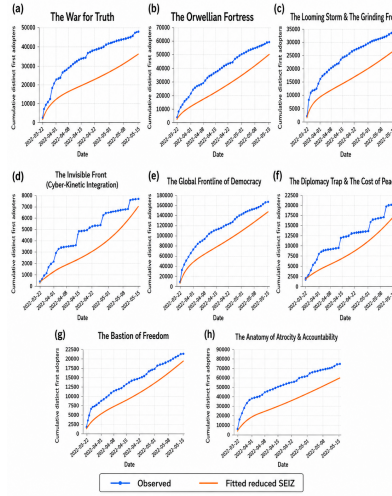


Fig. 2. Observed and fitted daily first-adopter incidence for selected SEIZ models, March 23–May 17, 2022, with 200-replicate bootstrap bands. Incidence is the estimated target; cumulative curves are derived by summation.

Implications for Information-Warfare Analysis. The stance decomposition provides insight into how political framing is distributed across narrative environments without identifying the actors or coordination mechanisms responsible for that distribution. Pro-Russia-aligned first adopters are most visible in *The Orwellian Fortress* and *The Diplomacy Trap*, suggesting that information-control and negotiation frames were comparatively more receptive to Pro-Russia-aligned content in the observed Twitter corpus. By contrast, Pro-Russia-aligned activity is limited in the atrocity and cyber narratives.

The larger Pro-Ukraine-aligned transmission point estimates across the eight narratives indicate stronger fitted diffusion for that stance under the present model and labeling procedure. Because the required joint reliability conditions are not met for formal stance leadership, this pattern is interpreted descriptively. It does not establish coordination, state sponsorship, campaign intent, or the operation of specific amplification networks. Rather, the results identify the narrative settings in which different political positions were more or less visible and more or less persistent during the observed period.

SEIZ vs. SEI_{AIDZ} . The two models describe different levels of the same diffusion process. SEIZ captures the aggregate spread of each narrative without separating stance, whereas the stance-stratified model shows how the Pro-Ukraine, Pro-Russia, and Neutral streams contribute to that aggregate pattern. Because the models have different compartment structures, their β and R_0 values are not directly comparable in magnitude.

For example, *The Invisible Front* has the largest aggregate SEIZ β and also the largest coupled R_0 . The stance-stratified results show that this pattern is

Table 3. Stance-specific $SEI_{AID}Z$ results by narrative

Narrative	First adopters PU/PR/N	Share (%) PU/PR/N	Stance β [95% CI] PU/PR/N	Counterfactual stance R_0 [95% CI] PU/PR/N	Coupled R_0 [95% CI]	Holdout RMSE ODE/Base
Global Frontline of Democracy	PU: 91,275 PR: 9,830 N: 62,247	PU: 55.88 PR: 6.02 N: 38.11	PU: 0.0629 [0.0340, 0.0829] PR: 0.0132 [0.0085, 0.0221] N: 0.0333 [0.0240, 0.0513]	PU: 0.868 [0.380, 1.119] PR: 0.210 [0.142, 0.317] N: 0.858 [0.594, 1.125]	1.505 [1.187, 1.821]	PU: 189.7/215.0 PR: 443.4/428.4 N: 445.3/455.7
Atrocity & Accountability	PU: 60,602 PR: 391 N: 15,984	PU: 78.73 PR: 0.51 N: 20.76	PU: 0.0399 [0.0277, 0.0590] PR: 0.0002 [0.0000, 0.0003] N: 0.0124 [0.0080, 0.0195]	PU: 0.505 [0.381, 0.795] PR: 0.002 [0.000, 0.005] N: 0.293 [0.200, 0.436]	0.658 [0.511, 0.962]	PU: 115.3/128.6 PR: 1.6/1.3 N: 104.7/75.5
Looming Storm & Grinding Front	PU: 19,141 PR: 809 N: 12,704	PU: 58.62 PR: 2.48 N: 38.90	PU: 0.0392 [0.0107, 0.0853] PR: 0.0040 [0.0014, 0.0090] N: 0.0070 [0.0000, 0.0152]	PU: 0.443 [0.302, 0.769] PR: 0.056 [0.020, 0.114] N: 0.158 [0.000, 0.386]	0.552 [0.282, 0.948]	PU: 133.9/185.1 PR: 37.3/40.6 N: 102.3/71.1
Oswellian Fortress	PU: 37,502 PR: 9,154 N: 12,738	PU: 63.14 PR: 15.41 N: 21.45	PU: 0.0424 [0.0244, 0.0663] PR: 0.0119 [0.0091, 0.0173] N: 0.0087 [0.0025, 0.0145]	PU: 0.617 [0.297, 0.826] PR: 0.200 [0.151, 0.267] N: 0.236 [0.072, 0.405]	0.838 [0.468, 1.080]	PU: 284.2/283.7 PR: 58.4/56.8 N: 142.4/142.3
War for Truth	PU: 25,003 PR: 2,396 N: 11,701	PU: 67.96 PR: 5.45 N: 26.59	PU: 0.0366 [0.0207, 0.0580] PR: 0.0037 [0.0019, 0.0065] N: 0.0128 [0.0065, 0.0254]	PU: 0.539 [0.258, 0.769] PR: 0.064 [0.032, 0.107] N: 0.362 [0.180, 0.665]	0.769 [0.508, 1.077]	PU: 306.1/309.8 PR: 19.0/20.8 N: 91.6/61.8
Diplomacy Trap & Cost of Peace	PU: 11,325 PR: 1,793 N: 6,974	PU: 56.37 PR: 4.92 N: 34.71	PU: 0.0547 [0.0300, 0.0962] PR: 0.0134 [0.0081, 0.0262] N: 0.0222 [0.0128, 0.0324]	PU: 0.843 [0.502, 1.116] PR: 0.232 [0.137, 0.363] N: 0.624 [0.331, 0.894]	1.302 [0.910, 1.686]	PU: 208.4/202.7 PR: 33.9/66.7 N: 79.0/253.6
Bastion of Freedom	PU: 17,701 PR: 419 N: 2,223	PU: 87.01 PR: 2.06 N: 10.93	PU: 0.0508 [0.0380, 0.0809] PR: 0.0013 [0.0003, 0.0023] N: 0.0116 [0.0083, 0.0176]	PU: 0.836 [0.546, 1.096] PR: 0.023 [0.005, 0.040] N: 0.325 [0.240, 0.448]	1.000 [0.719, 1.266]	PU: 100.4/104.4 PR: 9.4/9.2 N: 39.2/18.1
Invisible Front (Cyber-Kinetic)	PU: 4,897 PR: 22 N: 2,963	PU: 62.13 PR: 0.28 N: 37.59	PU: 0.0895 [0.0582, 0.1247] PR: 0.0005 [0.0002, 0.0011] N: 0.0195 [0.0110, 0.0311]	PU: 1.347 [0.861, 2.359] PR: 0.009 [0.002, 0.017] N: 0.540 [0.305, 0.803]	1.588 [1.143, 2.576]	PU: 195.3/202.6 PR: 0.6/0.5 N: 24.1/25.3

Notes: PU, PR, and N denote Pro-Ukraine, Pro-Russia, and Neutral. Estimates use daily first-time adoption over the 56-day window. Counterfactual stance R_0 isolates one transmission channel, whereas coupled R_0 retains all three. Holdout RMSE compares the ODE fit with a seven-day moving-average baseline over 11 days.

associated mainly with the Pro-Ukraine stream. *The Global Frontline of Democracy* also has a high coupled R_0 , but its diffusion is associated with both the Pro-Ukraine and Neutral streams. Similarly, *The Diplomacy Trap* has relatively large estimates under both models, while its stance-specific results show nonzero estimated transmission across all three streams, with the Pro-Ukraine component remaining the largest.

Together, the models show that a narrative can spread strongly in the aggregate while the stance-stratified model identifies which stance streams are most closely associated with that diffusion.

5 Conclusion

Using daily first-time unique-user adoption, this study combined calibrated SEIZ and a stance-stratified $SEI_{AID}Z$ model to examine eight conflict narratives. The results show that narratives differ not only in their aggregate diffusion patterns, but also in the stance streams most closely associated with that diffusion.

In the aggregate analysis, *The Invisible Front* had the largest numerical SEIZ β , while *The Diplomacy Trap* and *The War for Truth* showed the most consistent aggregate model performance. In the stance-aware analysis, Pro-Ukraine-aligned transmission estimates exceeded the corresponding Pro-Russia estimates across all narratives, with Neutral diffusion also playing an important role in several cases.

The main contribution is a framework that moves beyond identifying narrative themes to examine how strongly they spread and how their diffusion differs by stance. The findings describe modeled patterns in the observed Twitter corpus and should not be interpreted as evidence of persuasion, coordination, sponsorship, or political intent.

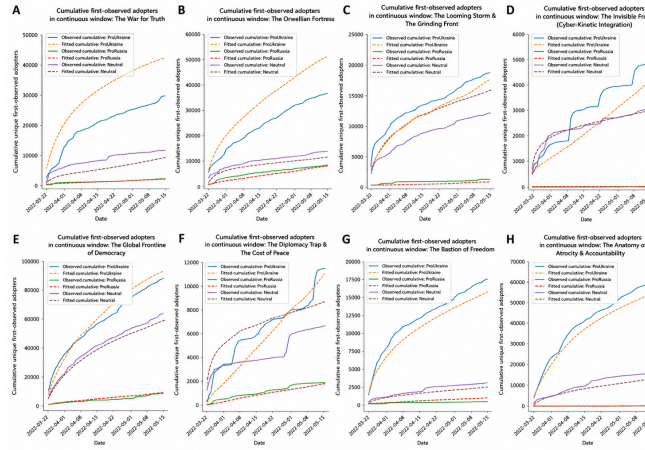


Fig. 3. Observed and fitted daily first-adopter incidence by stance, stance-stratified SEI_{ADZ} , with bootstrap bands. Separate β_A , β_D , and β_Z values govern the three stance streams while they compete for one susceptible population.

Limitations and Future Work. The analysis is restricted to Twitter and the continuous March 23–May 17, 2022 window; first observed adoption may not represent true first exposure, and the data may be affected by bots, deletion, coordination, and incomplete observation. Results also depend on the observed-user definition of the susceptible population, stance classification error, fixed transition rates, and pre-window exposure shares. Future work will consider longer observation windows, additional platforms, bot and coordination analyses, time-varying transmission, and models of observed stance switching.

Acknowledgments. This research is funded in part by the U.S. National Science Foundation (OIA-1946391, OIA-1920920), U.S. Army Research Office (W911NF-23-1-0011, W911NF-24-1-0078, W911NF-25-1-0147, W911NF-26-1-A163), U.S. Office of Naval Research (N00014-21-1-2121, N00014-21-1-2765, N00014-22-1-2318), U.S. Air Force Office of Scientific Research (FA9550-22-1-0332), U.S. Air Force Research Laboratory, U.S. Defense Advanced Research Projects Agency, the Australian Department of Defense Strategic Policy Grants Program, Arkansas Research Alliance, the Jerry L. Maulden/Entergy Endowment, and the Donaghey Foundation at the University of Arkansas at Little Rock. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the funding organizations.

References

1. Agarwal, N.: Developing a stance-induced epidemiological model to examine polarized information contagion. Sydney, Australia (2025)

2. Agarwal, N.: How do competing narratives spread? a stance-based epidemiological approach. Niagara Falls, Ontario, Canada (2025)
3. Agarwal, N., Al-Taweel, A.: Are narratives contagious? modeling narrative diffusion using epidemiological theories. Calabria, Italy (2024). https://doi.org/10.1007/978-3-031-78554-2_20
4. ALDayel, A., Magdy, W.: Stance detection on social media: State of the art and trends. *Information Processing & Management* **58**(4), 102597 (2021). <https://doi.org/10.1016/j.ipm.2021.102597>
5. Alieva, I., Kloos, I., Carley, K.M.: Analyzing russia’s propaganda tactics on twitter using network analysis and nlp. *EPJ Data Science* **13**(42) (2024). <https://doi.org/10.1140/epjds/s13688-024-00479-w>
6. Bawa, A., Kursuncu, U., Achilov, D., Shalin, V.L., Agarwal, N., Akbas, E.: Telegram as a battlefield: Kremlin-related communications during the russia-ukraine conflict. In: *ICWSM*. vol. 19, pp. 2361–2370 (2025). <https://doi.org/10.1609/icwsml.v19i1.35939>
7. Blei, D.M., Ng, A.Y., Jordan, M.I.: Latent dirichlet allocation. *Journal of Machine Learning Research* **3**(Jan), 993–1022 (2003)
8. Gemma Team: Gemma 4 technical report (2026)
9. Gilardi, F., Alizadeh, M., Kubli, M.: Chatgpt outperforms crowd workers for text-annotation tasks. *PNAS* **120**(30), e2305016120 (2023)
10. Google DeepMind: Gemma 4 model card (2026), https://ai.google.dev/gemma/docs/core/model_card_4, accessed 21 July 2026
11. Grootendorst, M.: Bertopic: Neural topic modeling with a class-based tf-idf procedure. arXiv preprint arXiv:2203.05794 (2022)
12. Gurung, M.I., Agarwal, N., Addai, E.: Modeling polarized information diffusion with sei (a) i (d) z: a stance-based epidemiological approach. *Social Network Analysis and Mining* **15**(1), 60 (2025)
13. Jin, F., Dougherty, E., Saraf, P., Cao, Y., Ramakrishnan, N.: Epidemiological modeling of news and rumors on twitter. In: *Proceedings of the 7th Workshop on Social Network Mining and Analysis* (2013)
14. Lai, C., Toriumi, F., Yoshida, M.: A multilingual analysis of pro-russian misinformation on twitter during the russian invasion of ukraine. *Scientific Reports* **14** (2024). <https://doi.org/10.1038/s41598-024-68870-1>
15. Maleki, M., Agarwal, N.: A comparative evaluation of the sir and seis epidemiological models to describe the diffusion characteristics of covid-19 polarizing viewpoints on online social networks. In: *HICSS* (2025)